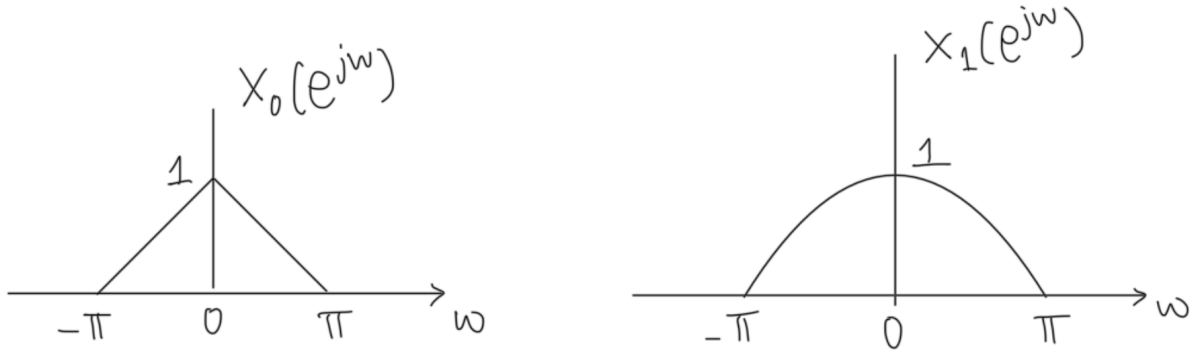
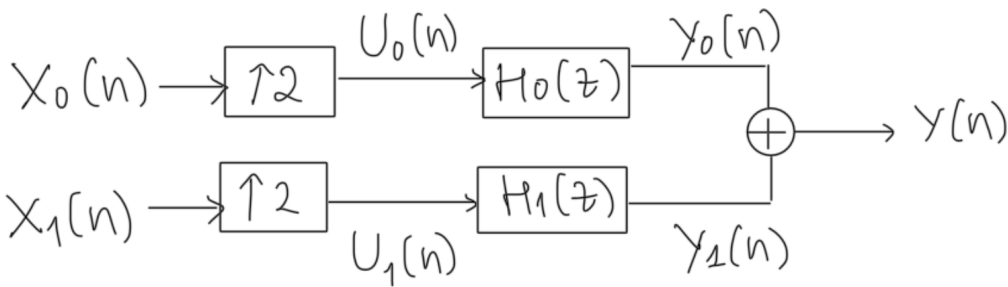


Homework 2

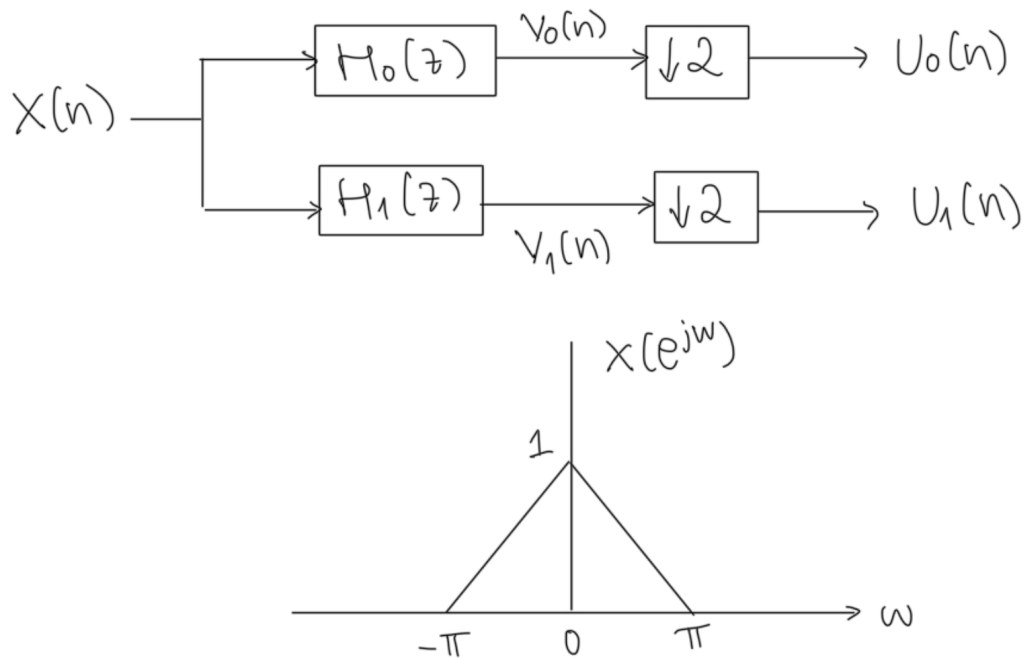
ECE 253 - Fall 2025

1. Consider the following system:



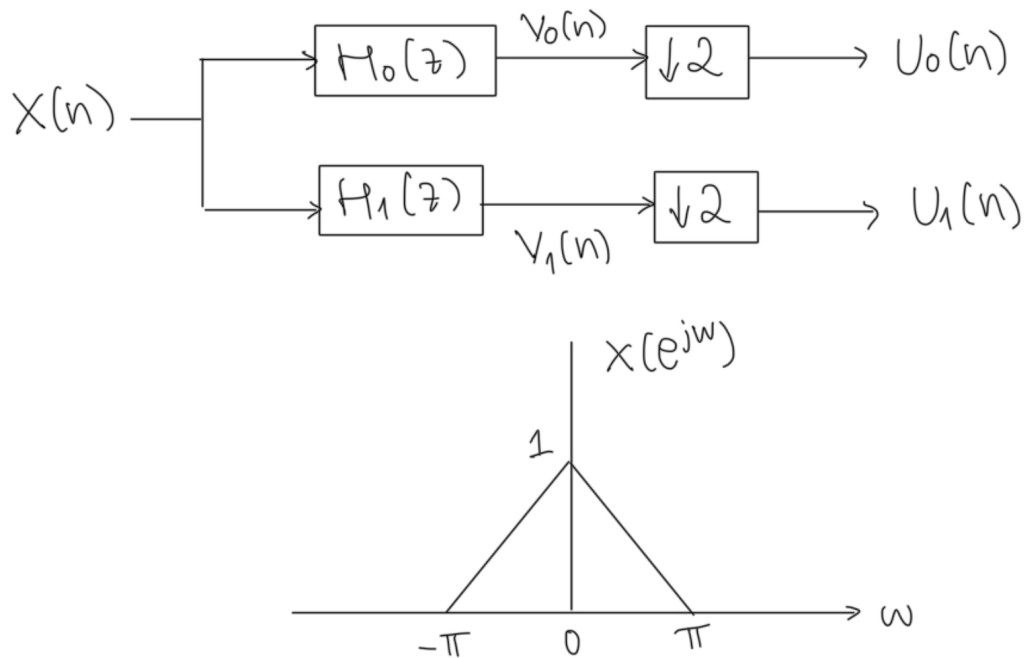
- Express $Y_0(z)$, $Y_1(z)$ in terms of $X_0(z)$, $H_0(z)$, $H_1(z)$.
- Assume that $H_0(e^{j\omega})$ is an ideal lowpass filter with cutoff frequency at $\frac{\pi}{2}$ and $H_1(e^{j\omega})$ is an ideal highpass filter with cutoff frequency at $\frac{\pi}{2}$. Sketch $U_0(e^{j\omega})$ and $U_1(e^{j\omega})$.
- Sketch $Y_0(e^{j\omega})$, $Y_1(e^{j\omega})$, $Y(e^{j\omega})$.
- Explain the operation of this system.

2. Consider the following system:



- Express $V_0(z)$, $V_1(z)$ in terms of $H_0(z)$, $H_1(z)$ and $X(z)$.
- Find the expression for $U_0(z)$, $U_1(z)$.
- Using the same filters H_0 and H_1 in Problem 1, sketch $U_0(e^{j\omega})$ and $U_1(e^{j\omega})$.
- Explain the operation of this system.

3. Consider the system similar to Problem 2.



- a) Assume that the filter $H_0(e^{j\omega})$ is an ideal low-pass filter with cutoff frequency $\frac{2\pi}{3}$, and $H_1(e^{j\omega})$ is an ideal high-pass filter with cutoff frequency $\frac{2\pi}{3}$. Sketch the frequency spectrum of $U_0(e^{j\omega})$ and $U_1(e^{j\omega})$ after downsampling.
- b) Compare $U_0(e^{j\omega})$ from part a with $U_0(e^{j\omega})$ in Problem 2. Discuss how aliasing appears in each case.